

**Vidya Vikas Institute of Engineering and Technology,
Mysore
Library and Information Centre**



VTU Question Papers

“Master of Computer Application”

II & IV Semester

June/July 2023

LIBRARY AND INFORMATION CENTRE

**Vidya Vikas Institute of Engineering and Technology
Bannur road, Alanahalli, Mysore-570028**

INDEX

S/L	Subject Code	Subject Title	Exam Date
1	22MCA21	Database Management System	June /July 2023
2	22MCA22	Object Oriented Programming Using Java	June /July 2023
3	22MCA23	Software Engineering	June /July 2023
4	22MCA24	Web Technologies	June /July 2023
5	22MCA261	Cryptography and Network Security	June /July 2023
6	20MCA41	Advances in Web Technologies	June /July 2023
7	20MCA42	Programming using C#	June /July 2023

--	--	--	--	--	--	--	--	--	--

Second Semester MCA Degree Examination, June/July 2023
Database Management System

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the Database System Environment with neat diagram. (10 Marks)
 b. Discuss the characteristics and advantages of Database Approaches. (10 Marks)

OR

- 2 a. Explain with proper diagram, the 3 – schema architecture of DBMS. (10 Marks)
 b. What are the different types of attributes? Explain with example. (10 Marks)

Module-2

- 3 a. Explain Unary Operation SELECT (σ) and prove it is commutative. (10 Marks)
 b. Explain Schema Update Operations, with a suitable examples. (10 Marks)

OR

- 4 a. With a suitable example, explain Join and division operation in relational algebra. (10 Marks)
 b. Explain in detail ER – to – Relational Mapping algorithm. (10 Marks)

Module-3

- 5 a. Explain with suitable example the basic structure of SQL query. (10 Marks)
 b. What are Views in SQL? Explain. (10 Marks)

OR

- 6 a. In SQL how to handle the Aggregate functions with group by and having clauses? With examples. (06 Marks)
 b. What are Aggregate functions? Explain with an examples. (06 Marks)
 c. Explain the architecture of JDBC main components and types of drivers. (08 Marks)

Module-4

- 7 a. Discuss informal design guidelines for relational schema. (10 Marks)
 b. What is Normalization? What are its advantages? Discuss 1NF, 2NF and 3NF. (10 Marks)

OR

- 8 a. Explain with an example the Boyce – Codd Normal Form (BCNF). (10 Marks)
 b. Discuss the different inference rules for functional dependencies. (10 Marks)

Module-5

- 9 a. Explain ACID properties of transaction in details. (10 Marks)
 b. Discuss the characterizing schedules based on recoverability. (10 Marks)

OR

- 10 a. Discuss a Lock – based concurrency control issue in DBMS transaction processing. (10 Marks)
 b. Describe Granularity of data items and Multiple Granularity locking. (10 Marks)

* * * * *

USN

--	--	--	--	--	--	--	--	--	--

22MCA22

Second Semester MCA Degree Examination, June/July 2023
Object Oriented Programming Using Java

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	What is scope and life time of a variable? Discuss each with the help of a java program.	10	L2	CO1
	b.	With the help of a proper java code, explain type conversion and type casting.	10	L2	CO1
OR					
Q.2	a.	Explain the working of for-each version of for loop using a java program.	10	L2	CO1
	b.	Write a java program which demonstrates constructor overloading and method overloading.	10	L3	CO2
Module – 2					
Q.3	a.	What is inheritance? Write a java program that implements simple inheritance.	10	L3	CO4
	b.	What are the two different user of super keyword in java? Illustrate each with proper examples.	10	L2	CO4
OR					
Q.4	a.	What is dynamic method dispatch? Explain how a superclass reference variable can refer to a subclass object with the help of a program.	10	L2	CO3
	b.	List out the conditions that are need to be followed while using while using abstract classes. Demonstrate the same by creating an abstract class and method.	10	L2	CO3
Module – 3					
Q.5	a.	Define interface. Discuss the features of interface and explain them with the help of a java program that implements an interface.	10	L2	CO3
	b.	List out the differences between abstract class and interface.	10	L1	CO3
OR					
Q.6	a.	What is package in Java? List and explain the system packages in java.	10	L2	CO1
	b.	If user wants to group all the similar type of classes and interfaces and keep them in a package and access them, how it can be done? Explain.	10	L3	CO1
1 of 2					

Module – 4

Q.7	a.	What is the use of multiple catch statement in exception handling? Discuss with a Java program.	10	L2	CO5
	b.	Whether try block can be nested in Java? If yes, demonstrate with the help of a java program.	10	L2	CO5

OR

Q.8	a.	Write a java program which uses throws keyword for handling exception.	10	L3	CO5
	b.	How to create a custom exception class in java? Demonstrate using a java program.	10	L3	CO5

Module – 5

Q.9	a.	Write a Java applet program which handles keyboard event.	10	L3	CO6
	b.	Explain the methods involved in life cycle of an applet.	10	L2	CO6

OR

Q.10	a.	How JButton class is used in swings? Explain.	10	L2	CO7
	b.	Write a Java program to display a frame using JFrame class.	10	L3	CO7

* * * * *

--	--	--	--	--	--	--	--	--	--

Second Semester MCA Degree Examination, June/July 2023

Software Engineering

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Describe software engineering code of ethics and professional practices as defined by IEEE/ACM.	10	L2	CO1
	b.	Why the software engineering is important? List the reasons. Brief the essential attributes of good software.	10	L2	CO1
OR					
Q.2	a.	Describe the waterfall and incremental software process models with suitable diagram.	10	L2	CO1
	b.	Discuss the principles of Agile methods.	05	L2	CO1
	c.	Explain the extreme programming release cycle.	05	L2	CO1
Module – 2					
Q.3	a.	Explain the classification of non-functional requirement with neat sketch and example.	10	L1	CO2
	b.	Explain the notations used in writing the software requirement specifications.	10	L1	CO2
OR					
Q.4	a.	Discuss the various difficulties that a software engineer faces during the eliciting and understanding requirements.	10	L1	CO2
	b.	Discuss the important activities of requirements engineering process with neat diagram.	10	L1	CO2
Module – 3					
Q.5	a.	Explain the generalization and inheritance with examples.	10	L1	CO3
	b.	Discuss about navigation of class models with suitable diagram and examples.	10	L1	CO3
OR					
Q.6	a.	What is N-array association? Illustrate the aggregation with associations and compositions with suitable examples.	10	L1	CO3
	b.	Explain the concept of reification and constraints with neat diagram and examples.	10	L1	CO3

Module – 4					
Q.7	a.	Explain system models with suitable example.	10	L2	CO4
	b.	With neat diagram, explain the working procedure of RUP with its advantages.	10	L2	CO4
OR					
Q.8	a.	Define design pattern. Explain the essential elements of design patterns.	10	L2	CO4
	b.	Explain in detail about the implementation issues involved in software engineering.	10	L2	CO4
Module – 5					
Q.9	a.	Discuss “Test Driven Development” (TDD) with its process and list out its benefits.	10	L4	CO5
	b.	Explain software evolution process with neat diagram.	10	L4	CO5
OR					
Q.10	a.	Describe the three main types of software maintenance. List of some difficulties and distinguishes between them.	10	L4	CO5
	b.	Explain why problems with support software might mean an organization has to replace legacy systems.	10	L4	CO5

* * * * *

--	--	--	--	--	--	--	--	--	--

Second Semester MCA Degree Examination, June/July 2023

Web Technologies

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Define HTTP. Explain the different phases of HTTP.	10	L2	CO6
	b.	Briefly explain the following: (i) URL (ii) MIME (iii) Web server (iv) Web browser	10	L2	CO6
OR					
Q.2	a.	Discuss the basic structure of a HTML5 webpage.	04	L2	CO6
	b.	List and explain any three from elements in HTML5 with a suitable example.	06	L2	CO6
	c.	Explain the following tags with examples: (i) Heading tag (ii) Hypertext link tag (iii) Image tag (iv) Audio and video tags (v) Progress tag	10	L2	CO6
Module – 2					
Q.3	a.	What is CSS? Describe the different levels of CSS style sheets and their precedence.	10	L2	CO2
	b.	Mention any five CSS selectors and explain their use with a suitable example.	10	L2	CO2
OR					
Q.4	a.	What is an array in Javascript? Explain the various ways of creating arrays. Mention any five array methods and explain their use.	10	L2	CO2
	b.	Write a Javascript program that accepts a value 'n' as input from the user and display the first 'n' fibonacci numbers as output.	04	L2	CO2
	c.	Discuss the following Javascript methods: (i) alert() (ii) prompt() (iii) confirm ()	06	L2	CO2
Module – 3					
Q.5	a.	What is boot strap? What are the features of boot strap? Create a simple webpage using boot strap.	10	L3	CO4
	b.	Discuss on the following by using boot strap code snippet: (i) Table (ii) Image (iii) Button (iv) Progress bar	10	L3	CO4

OR					
Q.6	a.	List out the various types of forms in boot strap. Explain their use with code snippet.	10	L3	CO4
	b.	Discuss briefly on the Grid system of boot strap.	05	L3	CO4
	c.	Explain boot strap progress bars with code snippet.	05	L3	CO4
Module – 4					
Q.7	a.	What is jQuery? What are the advantages of jQuery? Explain the syntax of jQuery script with a suitable example.	10	L2	CO1
	b.	Develop jQuery programs to implement the following jQuery effects: (i) Show() and hide () (ii) fadeIn() and fadeout()	10	L2	CO1
OR					
Q.8	a.	What is jQuery HTML? What are the methods used for DOM manipulation? Develop a jQuery program to get attribute values.	10	L2	CO1
	b.	What is an event? List the common events found in jQuery. Develop a jQuery program to implement mouse enter() jQuery event.	10	L2	CO1
Module – 5					
Q.9	a.	What is Angular JS? Explain the following Angular JS directives: (i) ng_app (ii) ng_model (iii) ng_bind	08	L2	CO5
	b.	Write an Angular JS program to use expressions.	06	L2	CO5
	c.	Briefly discuss the use of filter in Angular JS with an example.	06	L2	CO5
OR					
Q.10	a.	What is a Angular JS Service? Explain any three of them by using code snippet.	10	L2	CO5
	b.	Write an Angular JS program to demonstrate client-side form validation.	10	L2	CO5

* * * * *

USN

--	--	--	--	--	--	--	--	--	--

22MCA261

Second Semester MCA Degree Examination, June/July 2023

Cryptography and Network Security

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Explain different cipher methods for plaintext cryptography, obtain the ciphertext for substitution cipher 1 using 4 characters.	10	L2	CO1
	b.	Define security services. Explain the different categories and services of security services.	10	L2	CO1
OR					
Q.2	a.	Explain any two substitution techniques with example.	10	L2	CO1
	b.	With neat diagram, explain a model for Network Security.	10	L2	CO1
Module – 2					
Q.3	a.	With neat diagram, explain classical Feistel Network.	10	L2	CO2
	b.	Explain Data Encryption Standard (DES) with example.	10	L2	CO2
OR					
Q.4	a.	Explain Block Cipher design principles.	10	L2	CO2
	b.	Describe in detail Diffie – Hellman key exchange algorithm.	10	L2	CO2
Module – 3					
Q.5	a.	i) Explain Euler's Theorem with example ii) Describe Fermat's Theorem.	10	L2	CO3
	b.	Explain Chinese Remainder Theorem solve the problem for Chinese remainder where $X \equiv 3 \pmod{4}$ $X \equiv 2 \pmod{2}$ $X \equiv 4 \pmod{5}$	10	L2	CO3
OR					
Q.6	a.	Perform encryption and decryption algorithm using RSA algorithm for the following : $p = 3 ; q = 11 ; e = 7 ; \mu = 5$ $p = 5 ; q = 11 ; e = 3 ; \mu = 9$	10	L2	CO3

	b.	List and explain principles of public key cryptosystems.	10	L2	CO3
Module – 4					
Q.7	a.	Explain Secure Hash Algorithm (SHA).	10	L2	CO4
	b.	With neat diagram explain Hash function.	10	L2	CO4
OR					
Q.8	a.	With neat diagram, explain Message Authentication Function.	10	L2	CO4
	b.	Explain the two security of MACS.	10	L2	CO4
Module – 5					
Q.9	a.	With neat diagram, explain Generic model of Digital signature process.	10	L2	CO5
	b.	With a neat diagram, explain DSS approach.	10	L2	CO5
OR					
Q.10	a.	Explain key distribution scenario for transport key control scheme.	10	L2	CO5
	b.	Explain the technique used for distribution of public keys.	10	L2	CO5

* * * * *

USN

--	--	--	--	--	--	--	--	--	--

20MCA41

Fourth Semester MCA Degree Examination, June/July 2023
Advances in Web Technologies

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain any six string functions in PHP. (06 Marks)
- b. Explain Cookies in PHP. (04 Marks)
- c. Construct a PHP program to read student data from an XML file and store it into MYSQL. Retrieve and display the details using search options. (10 Marks)

OR

- 2 a. Explain Logical internal structure of arrays in PHP. (06 Marks)
- b. Illustrate Session Tracking in PHP. (04 Marks)
- c. Build a PHP form to display text base , table , radio button , submit button and clear button using XML. (10 Marks)

Module-2

- 3 a. List and explain different string methods in Ruby. (10 Marks)
- b. Develop a program for generating dynamic documents in Ruby on Rails. (10 Marks)

OR

- 4 a. Demonstrate layouts in Rails. (08 Marks)
- b. Develop a program in Ruby to read list of names from the keyboard, convert them all to upper case letters and place in an array and display in a sorted format. (12 Marks)

Module-3

- 5 a. Discuss the difference between Traditional web application and Ajax model. (06 Marks)
- b. Describe the different HTTP status code with their message. (04 Marks)
- c. Build a program to send the data to the server using GET method in Ajax. (10 Marks)

OR

- 6 a. Explain the technology behind Ajax. (05 Marks)
- b. Create a program to send data to the server using POST method in Ajax. (10 Marks)
- c. Explain the principles of Ajax. (05 Marks)

Module-4

- 7 a. Create a web page using array of XMLHttpRequest request object. (08 Marks)
- b. Build a program to cancel pending request using fallback pattern. (08 Marks)
- c. Describe Predictive fetch pattern. (04 Marks)

OR

- 8 a. Create a program for New comment Notifier using periodic refresh. (08 Marks)
- b. Describe Periodic refresh pattern. (04 Marks)
- c. Build a program for Page preloading using predictive fetch. (08 Marks)

Module-5

- | | | |
|----------|--|-------------------|
| 9 | a. Explain Fluid Grid system, with an example. | (05 Marks) |
| | b. Create a table using bootstrap table classes. | (10 Marks) |
| | c. Explain Responsive design with example. | (05 Marks) |

OR

- | | | |
|-----------|--|-------------------|
| 10 | a. Create a form using Optional form layouts of Bootstrap. | (10 Marks) |
| | b. Explain Prepended appended Input controls with example. | (10 Marks) |

* * * * *

USN

--	--	--	--	--	--	--	--	--	--

20MCA42

Fourth Semester MCA Degree Examination, June/July 2023

Programming using C#

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. With neat diagram, explain the work flow that takes place between .NET Source code, .NET Compiler and the execution engine. (10 Marks)
- b. Discuss between managed code and unmanaged code. (06 Marks)
- c. Explain the role of Common Type System. (04 Marks)

OR

- 2 a. What are namespaces? List and explain the purpose of any three namespaces. (10 Marks)
- b. Write short notes on:
 - (i) JIT compiler
 - (ii) Windows Communication Foundation (10 Marks)

Module-2

- 3 a. Explain how to create an array of objects of the class with the help of a C# program. (10 Marks)
- b. Explain the concept of static methods and static data members with suitable examples. (10 Marks)

OR

- 4 a. Write a C# program to explain accessor and mutator properties used in encapsulation. (10 Marks)
- b. Explain the following with example:
 - (i) Abstract class and abstract methods
 - (ii) Compile time and runtime polymorphism (10 Marks)

Module-3

- 5 a. What are delegates? Explain the concept of multicast delegate with an example. (10 Marks)
- b. Write a C# program to calculate square of numbers using delegates. (10 Marks)

OR

- 6 a. Write a C# program using try, catch, finally to explain any predefined exceptions. (10 Marks)
- b. Explain the properties and methods of Data Reader and Data Adapter Class. (10 Marks)

Module-4

- 7 a. Explain various keyboard events in C# windows applications. (10 Marks)
- b. Discuss the architecture of WPF with a neat diagram. (10 Marks)

OR

- 8** a. Explain the following:
(i) XAML definition and elements (10 Marks)
(ii) WPF Core Types (10 Marks)
b. What is GUI? List and explain basic controls of GUI. (10 Marks)

Module-5

- 9** a. Explain the architecture of a three tier web based application with a neat diagram. (10 Marks)
b. Write steps in session tracking with http session state using cookies. (10 Marks)

OR

- 10** a. Explain the controls from AJAX control toolkit. (10 Marks)
b. Explain different validation controls with suitable example supported by ASP.NET. (10 Marks)

* * * * *